Identifiability of independent sampling units in publicly deposited metabolomics data: a repository-level audit and documentary assessment

Anas Enoch[1*]

[1*]Mohammed VI University of Sciences and Health (UM6SS), Casablanca, Morocco.

Corresponding author(s). E-mail(s): anas_nour@um5.ac.ma;

Abstract

Introduction. Public metabolomics repositories enable secondary analysis and external validation, but valid inference requires that independent sampling units be identifiable. Unique sample labels do not identify distinct participants. **Objectives.** To assess whether repository metadata and linked documentation establish participant-level sampling structure under prespecified, affirmative evidence rules. **Methods.** A metadata-only census of 3,582 public MetaboLights accessions; a secondary audit of the factor metadata of 104 previously selected Metabolomics Workbench studies; a randomised documentary assessment of 40 of 371 unresolved MetaboLights candidates, reading deposit metadata together with linked publications; and a pre-verification of 19 human deposits whose rows an automated quality-control rule had removed. Source inventories were frozen and signed before reading, grading was deterministic, extraction was AI-assisted, and the author reviewed a prespecified subset. **Results.** Of 797 human single-matrix MS or NMR studies reaching the census independence test, none met the prespecified criteria. No study in the randomised assessment met the strict one-sample-per-participant standard (0 of 40; one-sided 95% hypergeometric upper bound 25 of 371 candidates). Documentation described repeated sampling of the same participants in 9 of 40 studies (8 on the author's re-review) and cultured cell lines in 3. In 17 of the 19 quality-control-excluded deposits, other fields identified study samples rather than blanks, 16 of them participant specimens. The author's review of a prespecified subset disagreed with the AI extraction only in assigning stated counts to the selected assay, never on the strict result. **Conclusion.** Under prespecified documentary criteria, participant-level sampling independence was not established for any study examined. The findings concern

what documentation establishes, not the true independence of deposited samples. Deposition standards should require explicit participant-to-sample linkage so that dependence can be modelled in reuse and external validation.

Keywords: metabolomics, data reuse, external validation, metadata, MetaboLights, Metabolomics Workbench, pseudoreplication, reporting standards

1 Introduction

Public repositories have made thousands of metabolomics studies available for reuse. MetaboLights and the Metabolomics Workbench together hold several thousand studies, and their deposits are increasingly reused for method development, benchmarking and the evaluation of predictive models [1, 2]. A common form of reuse treats each deposited study as an independent data set and each deposited sample as an observation. That treatment is valid only if an analyst can tell which samples come from the same participant.

The requirement is not that every study contribute one sample per participant. Longitudinal and repeated-measures designs are legitimate and often informative, and they can support external validation when the dependence among a participant's samples is modelled, clustered or reduced to one prespecified observation per participant. What any valid reuse needs is that the independent observational units be identifiable: the deposit, or the documentation linked to it, must allow the analyst to determine how many participants contributed, which samples belong to which participant, and whether any participant contributed more than one sample to the analysed set. When this information is missing, a repeated-measures study is indistinguishable from a cross-sectional one, and analyses that treat rows as independent participants risk pseudoreplication [3, 4]. In multi-study external validation [5] the same problem arises one level up: studies that share participants are not independent validation units.

The consequences of undetected dependence are documented in other omics fields. In transcriptomics, hidden duplicate samples across public data sets are common, and in simulation they inflate apparent cross-study validation performance [6]. Sample-level misannotation is also widespread in public expression data [7]. Controlled-access genomic repositories address this structurally: dbGaP, for example, requires a subject–sample mapping file for molecular data and allows one subject to map to several samples [8].

For metabolomics repositories, metadata quality has been assessed in general terms. Compliance with the minimum reporting standards of the Metabolomics Standards Initiative is incomplete and uneven across repositories [9, 10], automated quality scores for MetaboLights metadata vary widely [11], sample identifiers in Metabolomics Workbench files do not always match across sections [12], and only a minority of public mass-spectrometry files meet controlled-vocabulary metadata templates [13]. These studies show that metadata are often incomplete. They do not measure the specific property that study-level reuse depends on: whether the deposited information identifies the independent sampling units.

Here we measure that property directly. We define, before examining any study, what counts as affirmative evidence that the participants and samples of a deposited assay can be identified, and we apply the definition in four complementary ways, each with its own denominator and selection procedure: (i) a census of all 3,582 public MetaboLights accessions available on the census date, using deposited metadata only; (ii) an audit of subject-level fields in the factor metadata of 104 human Metabolomics Workbench studies; (iii) a documentary assessment of a randomly selected tranche of 40 MetaboLights candidates, reading the deposit metadata together with the methods sections of the linked publications; and (iv) a pre-verification of 19 human MetaboLights deposits whose sample rows an automated quality-control rule had removed in full. Throughout, we distinguish evidence that independence is established by the available documentation from the true independence of the underlying samples: failure to establish the former is not evidence against the latter.

2 Methods

All four analyses were metadata- and documentation-only. No measured metabolite value, data matrix, raw or derived data file was opened at any stage. Each analysis was specified in a written protocol, implemented in code with an offline test suite, and bound by SHA-256 hashes in a lock file signed by the author before execution. The locks were not registered with an external registry. Protocols, code, locks and outputs are public (Declarations).

2.1 Target property and evidence standard

The unit of interest is a deposited assay: the set of biological samples measured on one analytical platform for one sample matrix. We ask whether the deposited metadata, or the documentation linked to the deposit, identify the independent sampling units of that assay. Three pieces of information are needed: whether the samples come from human participants; how many distinct participants contributed to the assay; and whether any participant contributed more than one sample (several time points or visits, pre- and post-intervention samples, several specimens or technical replicates entered as separate biological samples), or, failing that, a sample-to-participant mapping that would let an analyst model the dependence.

Evidence had to be affirmative. Unique sample or source labels were treated as bookkeeping, not as evidence of distinct participants. Counts had to be stated as integers, or as sums of stated integers for groups the source described as disjoint; counts derived from percentages or by subtraction were not accepted. An enrolment or recruitment figure was not accepted as the number of participants in an assay unless the text tied it to the analysed samples. We call independence established in this sense "documented". It is distinct from true independence: a study whose documentation is silent may still consist of independent participants.

2.2 MetaboLights census

Universe.

A separately locked enumeration audit compared the MetaboLights REST study list with the public FTP mirror on 25 October 2026. The REST list (3,584 accessions) was a subset of the mirror (3,596 folders); the 12 mirror-only accessions were recorded as unresolved and not screened. Two accessions used as development exemplars were excluded, leaving a screening universe of 3,582 accessions.

Access boundary.

For each accession the census requested the directory index and every investigation, sample and assay file it listed (`i_`, `s_`, `a_`). Metabolite assignment files, data folders, raw and derived data and redirect destinations were refused by an allow-list guard. Requests were journalled with their HTTP status, size and declared length.

Rules.

Studies were screened in a fixed order: organism (M1: human samples only), sample-sheet integrity and quality-control rows (M2: rows classified as blanks, pooled or reference material, or quality control were removed; a study with no biological rows left was unresolved), duplicate sample names (M3), a single sample matrix (M4), and the independence test (M5). An independent subject was defined as a human individual contributing exactly one biological sample to the chosen assay, and M5 had two affirmative routes. R1 required a person-identifier column on the source node (for example `Characteristics[Subject ID]`, `[Participant]`, `[Donor]` or `[Patient]`, by a pattern frozen in code), present on every biological row, with exactly one sample per identifier. R2 required two layers. Layer 1 classified the design from the study design descriptors and description: at least one cross-sectional or case-control term and no longitudinal, repeated-measures, crossover, follow-up or pre/post term. Layer 2 required a sentence in the study description matching one of four frozen templates and containing no negation or plural-sampling token. Templates T1 and T2 state one sample per participant explicitly ("one sample was collected from each participant"). Templates T3 and T4 state a number of distinct participants, accepted only if the integer equalled the number of biological sources exactly. Both routes also required one sample per source and at least 50 sources. The strict criterion (L1) counted R1 and T1/T2 evidence; a sensitivity count added T3/T4 evidence, because equal cardinalities do not prove a one-to-one mapping. Studies that failed M5 but had at least 50 distinct source labels with one sample each ("source-unique") or at least 50 samples with a degenerate source column were recorded as unresolved candidates for documentary assessment.

Execution.

The census ran in three locked invocations: a full run (code v0.1.6) and two disclosed retry passes for transport failures and a rescreen of studies whose sample rows had all been classified as quality control (v0.1.7 and v0.1.8). The final record for each accession is the latest one.

2.3 Secondary audit of Metabolomics Workbench subject fields

This analysis is secondary. It reuses 104 human Metabolomics Workbench studies that an earlier value-blind screen, built for a different research question (a cross-cohort transfer design that was subsequently closed), had retained on 17 October 2026. The screen and its outcome are documented in a hashed closure record; its criteria are summarised in Table 1. The 104 studies are those that passed the screen's human-species, untouched-data, size and single-structure rules (E1–E5); they are not a random or representative sample of the Workbench, and no inference is drawn about the repository as a whole.

For each of the 104 studies we read the `factors` table returned by the Workbench REST interface (from the screen's local cache; no network request was made) and recorded whether any factor key named a subject, patient, participant, individual, donor, person, volunteer or proband. The inference is restricted to this endpoint. Workbench mwTab files, which may contain an optional subject column, embed measured values and were not opened, so nothing is claimed about subject fields elsewhere in the Workbench metadata or in deposited files. A documentary route through study descriptions was not available from the endpoints used and was recorded as not assessable.

Table 1 Provenance of the 104 Metabolomics Workbench studies (from the closure record of the earlier screen; counts as recorded there)

Step	n
Public studies in the Workbench species and source lists, 17 October 2026	4,635
of which human	1,906
E1: not human, or no species record (excluded)	2,716 / 13
E2: studies previously opened by the authors' earlier work (excluded)	216
PRE / PRE-N: fewer than 60 reported metabolites / fewer than 50 samples (excluded)	966 / 501
Pass PRE and PRE-N	223
E4: analysis attribution unresolved (excluded)	20
E5: repeated measures / multiple matrices / $n < 50$ after attribution (excluded)	48 / 41 / 10
Retained for this audit: eligible 72, E6 (metabolite-class rule, not applied here) 28, ambiguous naming 4	**104**

2.4 Documentary assessment of a random tranche (T1)

Population and selection.

The census produced 371 unresolved MetaboLights candidates with at least 50 unique source labels (312) or at least 50 samples with a degenerate source column (59). A tranche of 40 was selected as follows. Each candidate was ranked by SHA-256 of a salt concatenated with its accession, and the first 40 were taken. The salt was the output value of a NIST Randomness Beacon (version 2) pulse, at a time fixed in the

signed lock before the pulse existed (pulse 2-2005958, 30 October 2026, 15:00 UTC). No candidate was replaced. The tranche is therefore a simple random sample of 40 of the 371 candidates, and finite-population inference applies.

Sources.

Four sources were admissible: the deposit's investigation file (S1); publications named in the deposit by PubMed identifier or DOI, retrieved as full text from Europe PMC (S2); the first five publications, by first publication date, returned by a frozen Europe PMC full-text search for the exact accession string, usable only if their text contained the accession (S3); and author-supplied copies of S2 publications that were not open access, registered and hashed before the document freeze. No author was contacted.

Value-blind reading views.

Extractors read only a deterministic reading view of each publication: the title; the abstract with any Results, Findings, Conclusions or Discussion parts removed; methods-type sections; and participant-characteristics tables. Results and Discussion sections, figures and supplementary files were excluded. For PDF copies a heuristic view (front matter and the methods block) was used, and records citing it carry a flag.

Document freeze.

After retrieval, every document and view was listed with its hash in an evidence inventory, which the author signed before any evidence was read. No document could be added after the signature.

Questions and grades.

For the census-chosen assay of each study, extractors answered: Q1, whether the samples come from human participants; Q2, whether a source identifies the sample set analysed in the deposited assay; Q3, the number of distinct participants and whether it is scoped to the analysed set, to enrolment only, or absent; Q4, whether the source states one sample per participant, states repeated sampling, or is silent; Q6, overlap flags with other deposits; and Q7, whether a sample-to-participant mapping is published. Every answer other than "not stated" required a verbatim quotation with its location. A deterministic function assigned grades in a fixed order (Table 2). The STRICT grade, which requires a documented one-sample-per-participant structure, was designed for a downstream analysis that needs one sample per participant. It is more restrictive than the question of this paper: a study whose documentation permits an analyst to reconstruct the clustering of repeated samples would not reach STRICT. We therefore also report Q7 (a published sample-to-participant mapping) and the Q4 category, and we do not equate a zero STRICT count with the absence of reconstructible clustering.

Extraction, grading and adjudication.

Extraction was performed by AI sub-agents working under a written brief fixed and hashed before any record was produced (Section 2.8). The extractors were not told

Table 2 Grading rules for the documentary assessment, applied in order (protocol v0.2, §9)

Order	Condition	Grade
0	failed quotation check or schema error	invalid (returned)
1	assay membership (Q2) not established	UNRESOLVED
2	Q4 states repeated sampling, or Q7 shows repeated samples	CONTRADICTED
3	Q1: samples not from human participants	CONTRADICTED
4	Q1 not stated	UNRESOLVED
5	no participant count scoped to the analysed set	UNRESOLVED
6–7	stated count ≠ deposited biological rows	UNRESOLVED
8	count or assay link supported only by the deposit itself	UNRESOLVED
8b	assay link only from S3 papers without the accession in the quote	UNRESOLVED
9	one sample per participant stated in a publication (Q4 or Q7)	STRICT
10	otherwise	COUNT (sensitivity)

the deposits' biological row counts; the correspondence between a stated count and the deposited rows (rules 6–7) was computed only by the grading function. Every quotation was checked automatically for verbatim presence in the cited view. The author adjudicated every CONTRADICTED and STRICT record against the filed evidence; a rejected CONTRADICTED record became UNRESOLVED. No record was edited after grading.

2.5 Pre-verification of deposits removed by quality-control filtering

A diagnostic after the census identified 19 MetaboLights deposits with human organism values and at least 50 sample rows in which every row had been removed by the M2 quality-control rule. Because the census selected no assay for them, the four checks were applied to the biological samples listed in each deposit's sample file: V1, human participants; V2, the rows are identified as participants' samples by fields other than the sample-type label, or by a publication; V3, an integer number of biological samples stated in text (counting rows was not accepted); and V4, the sampling structure, failing on explicit repeated sampling. A study was verified if all four checks passed and failed if any check failed. Sources were the investigation, sample and assay files and named publications. The evidence set was frozen and signed before reading, extraction was blind to row counts, and quotations of sample names or individual-level rows were prohibited.

2.6 Statistical analysis

For the random tranche, the number of STRICT studies x among $n = 40$ drawn without replacement from $N = 371$ was used to bound the number K of documentation-recoverable candidates. The one-sided 95% upper bound is the largest K for which $P(X \leq x \mid N, K, n) > 0.05$ under the hypergeometric distribution; the lower bound is

defined symmetrically. Clopper–Pearson bounds scaled by N are reported for comparison. These are confidence bounds on the number of candidates whose documentation meets the standard, not bounds on the number of truly independent cohorts. All other results are reported as counts with their denominators.

2.7 Author review of the AI-assisted extraction

A two-part review was specified and hashed on 31 October 2026, before the reviewed studies were identified. The protocol intended an independent human reviewer; none could be recruited, and both components were performed by the author on 1–2 November 2026. The deviations from the protocol are stated here and in Section 3.4.

Component A (re-extraction).

The sampling frame was the 17 tranche studies with publication text that the author had not adjudicated. Ten were selected by a prespecified hash ranking of the accession strings. The author re-extracted Q3 and Q4 from the sealed sources using the frozen categories, without access to the original extractor's answers or quotations, but with prior knowledge of the overall study classifications and, in addition, after having seen the answers of a second AI model that had been run on the same ten studies in an earlier, abandoned attempt at this component (that sheet is preserved and labelled as an AI extraction). The author's sheet was assembled by script after the review, so per-study completion times were not recorded. Agreement is reported separately for Q3 scope, Q3 count (where both chose the assay-scoped category) and Q4, with every disagreement listed. The ten are not a random sample of the 40, and agreement is not extrapolated to the tranche.

Component B (contradiction review).

The author re-read the source texts of all 12 records finally graded CONTRADICTED and recorded, for each, whether an explicit statement supports the classification. Component B is not blind: the author had adjudicated the same records on 31 October 2026, so it is a consistency check by the same reader, not independent confirmation.

Neither component alters the frozen grades. Every quotation in both sheets was machine-checked for verbatim presence in the cited sealed file. The author's attestations, the sheets and the scoring notes are public.

2.8 Use of AI tools

Evidence extraction for the tranche and the pre-verification was performed by sub-agents of Claude (Anthropic), working under fixed, hashed written briefs. Each sub-agent read only the sealed files of its assigned studies, produced structured records with verbatim quotations, and had to pass an automated validator that checked the schema, the verbatim presence of every quotation and the form of every count. Grades were assigned by deterministic code, not by the sub-agents. The author adjudicated every CONTRADICTED record and carried out the human validation in Section 2.7. The briefs, validators and raw sub-agent outputs are public. Claude also assisted with code and drafting under the author's direction; the author is responsible for the content.

2.9 Integrity, access and data handling

The census, the Workbench audit and the documentary assessment were each executed under a signed lock that fixed the code, protocol and inputs by hash; code refused to run against any other version. Deviations and implementation limitations are recorded in the result records and summarised in the Results. Participant-level information was not recorded beyond counts. Pseudonymous source labels collected by the census are held in a checkpoint file that is not published. Full texts of publications are not redistributed.

3 Results

3.1 MetaboLights census

Of the 3,582 accessions, 3,569 reached a terminal status; 11 failed in transport and 2 returned persistent HTTP 403 responses (Table 3). Most exclusions were non-human studies (2,321). Of the human studies, 797 with a single sample matrix and an MS (714) or NMR (83) assay reached the independence test. **No study met the prespecified affirmative criteria under either the strict or the sensitivity count (L1 = 0).**

Neither route found qualifying evidence. Among the inspected records (the 658 studies whose source-node columns were tested against the R1 pattern and the 220 studies rescreened after all their rows had been classified as quality control), no qualifying person-identifier column was found. The R2 design layer passed in 18 of the 797 studies, and none of the 18 contained a qualifying one-sample-per-participant sentence. The census found 312 human studies with at least 50 distinct source labels and one sample per source, and 59 with at least 50 samples and a degenerate source column; these 371 formed the population for the documentary assessment.

In 220 studies every sample-sheet row was classified as quality control. In 183 of them the trigger was the value `blank` in a sample-type characteristic on every row. The census read the label as written; whether the rows are in fact biological samples was examined for the human subset in Section 3.5.

Table 3 Outcome of the MetaboLights census (3,582 accessions)

Status	Rule or reason	n
Excluded	non-human (M1)	2,321
Excluded	more than one matrix (M4)	67
Excluded	>1 sample per source or identifier (M5)	60
Excluded	no MS or NMR assay	1
Unresolved	source-unique, failed M5 (312 with ≥ 50 sources)	616
Unresolved	degenerate source column (59 with ≥ 50 samples)	121
Unresolved	M2: all rows QC / ambiguous QC label / unidentified rows	220 / 66 / 7
Unresolved	duplicate sample names / organism / organism part	13 / 3 / 1
Refused	format / structure / index	61 / 10 / 2
Not terminal	transport failure / HTTP 403	11 / 2
Eligible	strict and sensitivity L1	**0**

3.2 Secondary audit of Metabolomics Workbench subject fields

None of the 104 previously selected human studies had a factor key naming a subject, patient, participant, individual, donor, person, volunteer or proband in the `factors` table. The most frequent keys were design factors: sample source (21), treatment (16), group (13), sex or gender (13), diagnosis (8). Two limitations bear on this result. The inherited repeated-measures detector did not recognise underscore or camel-case time keys (for example `POST_SURGERY_TIME`, `collectionWeek`), so dependence in these studies is undercounted. Factor keys such as cell line, cell culture medium and post-mortem interval show that a "human" species label did not by itself establish human participants. These findings concern the `factors` endpoint of 104 selected studies only.

3.3 Documentary assessment of the random tranche

Of the 40 studies, 29 had at least one publication reading view in the frozen evidence set and 11 had none. Under the prespecified procedure, no qualifying statement that each participant contributed one sample to the analysed set was identified in any reading view, and no view contained a sample-to-participant table; Q7 was "none" for all 40 studies. **No study reached the STRICT grade ($x = 0$).** Because STRICT requires a stated one-sample-per-participant structure, this count does not measure how many studies' clustering could be reconstructed from other documentation; that question is addressed by Q7 and the Q4 categories reported below. The one-sided 95% hypergeometric upper bound on the number of the 371 candidates whose documentation meets the standard is 25 (Clopper–Pearson: 26.8).

Table 4 Final grades of the 40 tranche studies after adjudication

Grade (reason)	n
STRICT	0
COUNT (sensitivity only)	5
CONTRADICTED: repeated sampling in the chosen assay	9
CONTRADICTED: samples not from human participants (cell lines)	3
UNRESOLVED: adjudication rejected	3
UNRESOLVED: assay membership not established	6
UNRESOLVED: no assay-scoped participant count	5
UNRESOLVED: stated count below / above deposited rows	3 / 3
UNRESOLVED: count supported only by the deposit	3

Where documentation addressed sampling, it often described dependence. Nine studies (22.5%) documented repeated sampling of the same participants within the chosen assay (serial time points, pre/post-treatment samples, follow-up visits, several cysts per patient, up to 56 time points per participant), and three (7.5%) consisted of cultured cell lines rather than participant samples. Together these account for the 12 CONTRADICTED classifications after adjudication. All twelve had entered the candidate pool through unique or degenerate source labels. Under the author's re-review (Section 3.4) one of the nine repeated-sampling classifications is not explicitly supported, giving 8 of 40 (20%) as a sensitivity figure.

Stated counts frequently failed to reconcile with the deposits. Among the 13 studies not finally graded CONTRADICTED that reported an assay-scoped participant count with a publication source, 7 matched the number of deposited biological rows and 6 did not (3 lower, 3 higher). None of the 7 had one-sample-per-participant evidence. The candidate pool also contained four separately deposited ionisation-mode assays of one sample set (MTBLS1122, MTBLS1124, MTBLS1129, MTBLS1130), so the number of candidates overstates the number of distinct studies.

The result does not depend on unread documents or on adjudication. Three named publications could not be obtained; if all three had supplied STRICT evidence, x would be 3 and the upper bound 65. Adjudication can only confirm or reject CONTRADICTED and STRICT grades, and a rejection yields UNRESOLVED.

3.4 Author review of the AI-assisted extraction

Component A.

Q3 scope agreed in 6 of 10 studies; Q3 count agreed in 4 of 4 studies in which both the author and the extractor chose the assay-scoped category; Q4 agreed in 10 of 10 (both answered "not stated" for every study, a shared null that does not test sensitivity for detecting repeated sampling). The four Q3 disagreements (MTBLS718, 11406, 2774, 13787) all ran in one direction: the author declined to assign a stated count to the selected assay where the extractor had done so, because the documents did not say which cohort or sample set the deposit held, or gave a sample count rather than a participant count. In no case did the author find a count or a sampling statement the extractor had missed. Three of the four records were already UNRESOLVED; under the author's reading the fourth (MTBLS11406) would move from COUNT to UNRESOLVED, reducing the sensitivity-only STRICT + COUNT figure from 5 to 4. No disagreement bears on the STRICT count.

Component B.

Of the 12 CONTRADICTED records, 11 were confirmed by an explicit statement in the cited source (the same sentence the extractor had cited), none was disputed, and one (MTBLS10972, deposit text only) was judged unresolved: the deposit describes a post-meal time course but does not state that multiple specimens from the same individuals entered the deposited assay. The author had confirmed this record at adjudication while noting the evidence as weak. As a sensitivity figure, documented repeated sampling is 8 of 40 (20%) rather than 9 of 40, and the CONTRADICTED total 11 rather than 12; the three cell-line classifications are unchanged. The primary figures in Table 4 are the frozen ones.

3.5 Deposits removed by quality-control filtering

In 18 of the 19 human deposits whose rows had all been removed by the quality-control rule, every sample row carried `Characteristics[Sample type] = experimental blank`; in the remaining deposit, 94 of 95 rows did. In 17 of the 19, other deposited fields identified the rows as the study's biological samples rather than blanks (V2 passed): disease-group or treatment factor values on the same rows, sample-collection

protocols, or the study description. In 16 of these 17 the same fields describe participant specimens; in MTBLS15165 they describe stored platelet units whose donors are not mentioned, so the human origin of the material is not established (V1 not established). Table 5 lists the supporting fields per deposit. In these 17 the sample-type annotation conflicts with the other fields. Three deposits were verified (all four checks passed, with stated sample counts of 368, 144 and 85), eight failed (cell lines or non-human rows in two, pooled material in two, repeated sampling in four) and eight were unverifiable, mostly because no sample count was stated.

Two further inconsistencies were found. In both assay files of MTBLS14421, the extract, assay and normalisation names of 50 of 53 rows carry the label of a different sample, and in 30 rows that label belongs to the other study group (15 control rows carry patient labels and 15 patient rows carry control labels). This is a linkage inconsistency between the sample sheet and the assay table; the deposit does not show which label is correct, and we make no claim about the published analysis. MTBLS14358 and MTBLS14376 share a title and a cohort description; whether their sample sets overlap is not established from the deposits.

3.6 Implementation limitations

The frozen implementation had three limitations, each disclosed in the result records. One reading view included a differential-metabolite table whose caption used an abbreviation that escaped the exclusion pattern; nothing was cited from it. The PDF view heuristic did not recognise "STAR+METHODS" headings; no graded record depended on such a block. The count check accepted an integer from a deposit quotation when the participant-count question also cited a publication; in the one affected record the publication stated the disjoint group sizes that sum to the count.

4 Discussion

The question this study asks is not whether the original investigators sampled their participants acceptably, but whether a secondary analyst, working from the public deposit and its linked documentation, can reconstruct the sampling structure well enough to identify independent units and account for dependence. Four analyses with different denominators and different targets bear on it: the census and the documentary assessment tested a one-sample-per-participant criterion; the Workbench audit examined one metadata endpoint of 104 selected studies; the pre-verification examined sample classification, not participant linkage. None establishes a repository-wide prevalence of irreconstructible sampling structures. Together they give convergent evidence that the information needed to identify independent sampling units is frequently absent from deposited metadata and often absent from linked publications under an affirmative-evidence standard. We draw three implications.

Missing subject-level linkage prevents correct dependence modelling.

A repeated-measures design is not a defect; it becomes one for reuse only when the deposit does not say which samples share a participant. In the MetaboLights census no inspected record carried a person-identifier column, and in the 104 Workbench studies

Table 5 Deposits whose every sample row is labelled `experimental blank`, and the other deposited fields that identify the rows as study samples rather than blanks (V2). For MTBLS15165 the fields establish study material, not identifiable donors (V1 not established). Quotations are in the public evidence records.

Accession	V2	Fields supporting V2
MTBLS20	pass	investigation text (urine processing protocol); sample-file organism column
MTBLS148	pass	investigation text (urine collection from donors); sample-file factor values
MTBLS6260	pass	investigation text (plasma collection); sample-file time factors
MTBLS9021	pass	investigation text (baseline blood from patients); sample-file factor values
MTBLS11048	not established	none (cell-line rows)
MTBLS12684	pass	named publication; sample-file disease characteristic; investigation text
MTBLS12824	pass	investigation text; sample-file age-group and exercise factors
MTBLS13136	pass	investigation text; sample-file disease factor
MTBLS13319	not established	none (pooled plasma, method study)
MTBLS13382	pass	investigation text (DBS and serum collection); sample-file sample-type factor
MTBLS13634	pass	investigation text; sample-file treatment factor and group characteristic
MTBLS13827	pass	investigation text; sample-file treatment factor and group characteristic
MTBLS14358	pass	investigation text; sample-file organism column and group factor
MTBLS14376	pass	investigation text; sample-file group factor
MTBLS14421	pass	investigation text (serum from patients and volunteers); sample-file factor
MTBLS14426	pass	investigation text (urine and fasting serum collection); sample-file factor
MTBLS14865	pass	investigation text (urine before surgery or at examination); sample-file treatment factor
MTBLS15004	pass	investigation text (faecal samples from patients in two groups)
MTBLS15165	pass[†]	investigation text (stored platelet units); sample-file replicate factor

[†]Study material established; donors not described.

examined no factor key named a subject. The documentary tranche showed what this costs: 9 of 40 randomly selected studies (22.5%) documented repeated sampling of the same participants in the deposited assay (serial sampling, pre/post samples, several specimens per patient), and a further 3 (7.5%) consisted of cell lines rather than participants. For the nine, no qualifying participant-to-sample mapping was identified under the prespecified metadata and documentary procedures. An analyst treating these rows as independent would be pseudoreplicating [3, 6]; an analyst wanting to model the dependence correctly could not. The same gap applies at the study level, where separately deposited assays of one sample set (MTBLS1122 and its three ionisation-mode siblings; the two deposits sharing a cohort description in the pre-verification

set) would be counted as independent validation units by any procedure that counts accessions.

Automated metadata labels can conceal valid biological specimens.

The census removed every sample row of 220 studies as quality-control material because a sample-type field read `blank`. For the 19 human deposits we examined, other fields on the same rows (disease-group factors, collection protocols, a named publication) identified study samples rather than blanks in 17, and participant specimens in 16 of those. The label appears to be a template default rather than a description. The practical consequence runs in both directions: a screen that trusts the label discards usable studies, and a screen that ignores it cannot distinguish blanks from specimens. Only the depositor can resolve this, and the finding argues for vocabulary defaults that are empty rather than pre-filled. The linkage inconsistency in MTBLS14421, where assay rows carry the labels of other samples across study groups, is a different kind of defect: it is detectable from the deposit, but not resolvable from it.

Reproducible repository screening needs explicit evidence standards.

A screen that accepts unique sample names as evidence of independent participants would have passed hundreds of census studies; ours passed none, and the documentary tranche showed why that stricter standard was warranted, since 22.5% of the uniquely labelled candidates documented repeated sampling and 7.5% were not participant samples at all. Whatever standard a reuse study adopts, it should be stated before screening, applied by code that refuses other inputs, and reported with its failure categories, so that the gap between "documentation establishes independence" and "samples are independent" is visible rather than silently closed. Our results also show the limits of such screening: 11 of 40 randomly selected studies had no retrievable publication text, and the STRICT criterion of the documentary assessment, designed for an analysis needing one sample per participant, is stricter than the reconstructibility question of this paper. The zero STRICT count should not be read as zero reconstructible studies.

Limitations.

The randomised documentary assessment covers one repository, and its 40 studies were drawn from the 371 unresolved candidates that the census had already filtered (human, single-matrix, at least 50 labelled sources), not from all MetaboLights studies; its design-based bound refers to that population, and its point estimate is not the result. The Workbench analysis is secondary, restricted to one metadata endpoint and to studies selected for another purpose. Evidence extraction was AI-assisted. The author's review (Section 3.4) found no case in which the extractor missed a count or a sampling statement, and its disagreements all made the frozen records more, not less, conservative; but it was performed by the author, not an independent reviewer, after partial exposure to a second model's answers, and its Q4 agreement is agreement on a shared negative, which says nothing about sensitivity for detecting repeated sampling. Component B was a consistency check by the reader who had adjudicated the same records. The review therefore supports the zero-STRICT finding and the direction of

the classifications; it does not establish extraction accuracy against an independent human standard. The frozen implementations had the limitations listed in Section 3.6. Documentation can be missing for reasons unrelated to the study's design, and none of our results estimates how many deposited studies are in fact composed of independent participants.

Recommendations.

Repositories should require, at deposition, a participant identifier or a subject-to-sample mapping on every biological sample, as controlled-access genomic archives already do [8], and should flag deposits that share a sample set. Sample-type vocabularies should have no pre-filled default. Our recommendation concerns deposition metadata rather than manuscript reporting. The STROBE-MetEpi reporting guideline for metabolomic epidemiology, currently under review [14], already asks authors to report participant eligibility and sampling (item 5a), the biological sample collection protocol including timing and whether conditions were the same for every sample (6a), any independent validation population (17), the final number of individuals in the primary analyses (19a), the time points at which profiles were measured in cohort studies (21c), and repository access (31a). None of its 31 items asks for a machine-readable mapping from deposited samples to participants, and a manuscript statement that 100 participants contributed cannot be joined to deposited rows unless the deposit carries that mapping. A deposition-level linkage requirement would complement, not replace, this manuscript-level guidance. (The guideline's scope excludes methodological studies, so the present audit is not itself assessed against it.) None of this requires investigators to abandon repeated measurements; it requires that the deposit say where they are.

Declarations

- Funding: This research received no specific funding from any funding agency in the public, commercial or not-for-profit sectors.
- Competing interests: The author declares no competing interests.
- Use of AI tools: see Section 2.8.
- Data and code availability: protocols, code, locks, sealed evidence inventories, evidence records, grades and result records are public in the repository https://github.com/Anas-Enoch/Pathway_Identifiability (commits `b5cd507`, `3d376d0`, `195a60d`, `566af9d`). Full texts of publications are not redistributed; their SHA-256 values and retrieval URLs are recorded.
- Author contribution: A.E. designed the study, executed the protocols, adjudicated and reviewed the records, and wrote the manuscript.

References

[1] Haug, K., *et al.*: MetaboLights: a resource evolving in response to the needs of its scientific community. Nucleic Acids Research **48**(D1), 440–444 (2020) https://doi.org/10.1093/nar/gkz1019

[2] Sud, M., *et al.*: Metabolomics Workbench: an international repository for metabolomics data and metadata, metabolite standards, protocols, tutorials and training, and analysis tools. Nucleic Acids Research **44**(D1), 463–470 (2016) https://doi.org/10.1093/nar/gkv1042

[3] Hurlbert, S.H.: Pseudoreplication and the design of ecological field experiments. Ecological Monographs **54**(2), 187–211 (1984) https://doi.org/10.2307/1942661

[4] Lazic, S.E.: The problem of pseudoreplication in neuroscientific studies: is it affecting your analysis? BMC Neuroscience **11**, 5 (2010) https://doi.org/10.1186/1471-2202-11-5

[5] Riley, R.D., *et al.*: External validation of clinical prediction models using big datasets from e-health records or IPD meta-analysis: opportunities and challenges. BMJ **353**, 3140 (2016) https://doi.org/10.1136/bmj.i3140

[6] Waldron, L., Riester, M., Ramos, M., Parmigiani, G., Birrer, M.: The Doppelgänger effect: hidden duplicates in databases of transcriptome profiles. Journal of the National Cancer Institute **108**(11), 146 (2016)

[7] Toker, L., Feng, M., Pavlidis, P.: Whose sample is it anyway? Widespread misannotation of samples in transcriptomics studies. F1000Research **5**, 2103 (2016)

[8] National Center for Biotechnology Information: dbGaP submission guide. https://www.ncbi.nlm.nih.gov/gap/docs/submissionguide. Accessed 31 October 2026

[9] Spicer, R.A., Salek, R., Steinbeck, C.: Compliance with minimum information guidelines in public metabolomics repositories. Scientific Data **4**, 170137 (2017)

[10] Spicer, R.A., Salek, R., Steinbeck, C.: A decade after the metabolomics standards initiative it's time for a revision. Scientific Data **4**, 170138 (2017)

[11] Ferreira, J.D., Inácio, B., Salek, R.M., Couto, F.M.: Assessing public metabolomics metadata, towards improving quality. Journal of Integrative Bioinformatics **14**(4) (2017)

[12] Powell, C.D., Moseley, H.N.B.: The mwtab Python library for RESTful access and enhanced quality control, deposition, and curation of the Metabolomics Workbench data repository. Metabolites **11**, 163 (2021)

[13] Jarmusch, A.K., *et al.*: ReDU: a framework to find and reanalyze public mass spectrometry data. Nature Methods **17**, 901–904 (2020)

[14] Kelly, R.S., Reinke, S.N., et al.: A reporting guideline for observational studies in metabolomic epidemiology: explanation and elaboration of the STROBE-MetEpi checklist. Research Square preprint, 25 June 2026, under review (2026)